The norm–supernorm relation for the size ensemble

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Abstract

We prove that the product of the reciprocal norm and reciprocal supernorm partition sums of size n tends to one. The logarithmic derivative of the supernorm generating function gives a positive recurrence. Two successive bounds on its coefficients turn the known cumulative asymptotic into the required individual asymptotic.

Task 1: the original problem

For a partition λ of size $|\lambda| = n$, write

$$N(\lambda) = \prod_i \lambda_i, \qquad \widehat{N}(\lambda) = \prod_i p_{\lambda_i}, \qquad v_n = \sum_{|\lambda|=n} \frac{1}{N(\lambda)}, \qquad u_n = \sum_{|\lambda|=n} \frac{1}{\widehat{N}(\lambda)},$$

where p_j is the jth prime, $p_1 = 2$, and $u_0 = v_0 = 1$. The intended W_{size} in the source is the reciprocal norm sum v_n.

Theorem 1. *As $n \to \infty$,*

$$u_n \sim \frac{e^\gamma}{n}, \qquad \textit{and consequently} \qquad v_n u_n \longrightarrow 1.$$

The product assertion is Conjecture 3.8 in both versions of [1]. Its equivalent size-supernorm assertion is Conjecture 3.5 of the preprint and Conjecture 3.7 of the journal final. They are counted as one target. The separate perimeter conjecture is not used in this proof. We use three established inputs from [1]: the cumulative theorem, the prime number theorem, and Lehmer's reciprocal norm theorem:

$$C_n := \sum_{j=0}^{n} u_j \sim e^\gamma \log n, \qquad p_n \sim n \log n, \qquad v_n \sim e^{-\gamma} n. \tag{1}$$

See respectively Theorem 3.4, Lemma 4.4, and Theorem 1.1 there.

The logarithmic-derivative kernel

As an identity of formal power series,

$$F(z) = \sum_{n \geq 0} u_n z^n = \prod_{d \geq 1} (1 - z^d/p_d)^{-1}.$$

Every coefficient depends on finitely many factors. Formal logarithmic differentiation therefore gives

$$\frac{zF'(z)}{F(z)} = \sum_{k\geq 1} a_k z^k, \qquad a_k = \sum_{d\mid k} \frac{d}{p_d^{k/d}}. \tag{2}$$

Equating coefficients yields the exact positive recurrence

$$nu_n = \sum_{k=1}^{n} a_k u_{n-k}. \tag{3}$$

Lemma 1. *As $k \to \infty$,*

$$a_k = \frac{k}{p_k} + O(k^{-1}) \sim \frac{1}{\log k}.$$

Consequently a_k is bounded and $\sum_{k\leq x} a_k = O(x/\log x)$ as $x \to \infty$.

Proof. Write $R_k = a_k - k/p_k \geq 0$ and put $r = k/d \geq 2$ in its sum. For $2 \leq r \leq \sqrt{k}$, the elementary bound $p_d \geq d + 1 > d$ gives

$$\frac{d}{p_d^r} \leq \left(\frac{r}{k}\right)^{r-1}.$$

For $k \geq 4$, the $r = 2$ term is at most $2/k$. The terms with $3 \leq r \leq \sqrt{k}$, after dropping the divisibility restriction, total at most

$$\sum_{r=3}^{\infty} k^{-(r-1)/2} = \frac{1/k}{1 - 1/\sqrt{k}} \leq \frac{2}{k}.$$

For $r > \sqrt{k}$, the corresponding d are less than $\sqrt{k}$. There are at most $\sqrt{k}$ such integers, each term is at most $\sqrt{k}\, 2^{-\sqrt{k}}$, and thus

$$0 \leq R_k \leq \frac{4}{k} + k\, 2^{-\sqrt{k}} = O(k^{-1}). \tag{4}$$

The prime number theorem now proves the kernel asymptotic and boundedness. To bound its partial sums, the $k \leq \sqrt{x}$ terms total $O(\sqrt{x})$; the remaining terms are each $O(1/\log x)$, giving $O(x/\log x)$. □

Two bounds and the limiting constant

Boundedness of a_k and (1) first imply from (3)

$$nu_n = O(C_{n-1}) = O(\log n), \qquad u_n = O(\log n/n).$$

Split (3) at $k = n/2$. For $k \leq n/2$, this initial bound gives $u_{n-k} = O(\log n/n)$ uniformly, so Lemma 1 implies

$$\sum_{k\leq n/2} a_k u_{n-k} = O\left(\frac{\log n}{n}\right) O\left(\frac{n}{\log n}\right) = O(1).$$

For $k > n/2$, the kernel bound is uniformly $O(1/\log n)$, whence

$$\sum_{n/2<k\leq n} a_k u_{n-k} = O\left(\frac{1}{\log n}\right) C_{\lceil n/2\rceil - 1} = O(1).$$

Consequently

$$u_n = O(1/n). \tag{5}$$

Use (5) in the first half of the same recurrence to improve its contribution to

$$\sum_{k\le n/2} a_k u_{n-k} = O\left(\frac{1}{n}\right) O\left(\frac{n}{\log n}\right) = o(1). \tag{6}$$

In the second half, $a_k \log n \to 1$ uniformly for $n/2 < k \le n$: indeed $a_k \log k \to 1$ and $\log k/\log n \to 1$ uniformly there. Since every $u_j \ge 0$, this gives

$$\sum_{n/2<k\le n} a_k u_{n-k} = \frac{1+o(1)}{\log n} C_{\lceil n/2\rceil - 1} \longrightarrow e^{\gamma}.$$

Together with (6), recurrence (3) proves $nu_n \to e^{\gamma}$. Multiplication by Lehmer's asymptotic in (1) proves $v_n u_n \to 1$. □

Scope and verification

All positive integer parts and their multiplicities are unrestricted. The proof uses the already proved cumulative theorem without differentiating its error term. Formal coefficient operations require no boundary convergence. The companion verifier independently enumerates partitions, compares a product expansion with the exact recurrence, checks the kernel remainder bound with rational arithmetic, and supplies numerical examples. Finite checks are not a proof of the limit. Review is by the authoring agent only, without independent peer review or a proof assistant.

Proposed follow-up Tasks 2–5

Task 2: Obtain an explicit remainder estimate for $nu_n - e^{\gamma}$, with a stated constant and threshold.

Task 3: Prove a coefficient transfer theorem for positive recurrences of the form (3) when $a_k \sim \alpha/\log k$ and $\sum_{j\le n} u_j \sim c(\log n)^{\alpha}$, for $\alpha, c > 0$.

Task 4: Determine the leading constant for the reciprocal supernorm sum over partitions of size n with distinct parts.

Task 5: For each fixed $\beta > 1$, determine the leading asymptotic of $\sum_{|\lambda|=n} \widehat{N}(\lambda)^{-\beta}$, including its constant.

References

[1] J. C. Lagarias and C. Sun, *Asymptotics of reciprocal supernorm partition statistics*, The Ramanujan Journal **65** (2024), 487–520. Journal version (DOI). Preprint: arXiv:2308.15303v1.